AI Campus Assistant

Final Year Project Documentation

1. Abstract

AI Campus Assistant is a Retrieval-Augmented Generation (RAG) based intelligent system designed to provide smart responses to student queries using institutional documents. The system integrates Google Gemini AI with FAISS vector search to ensure contextual and accurate answers. It supports multilingual interaction, voice input/output, secure authentication, and an analytics-driven admin dashboard.

2. Problem Statement

Students often face difficulty accessing accurate academic information such as admission procedures, exam schedules, fee structures, and syllabus details. Manual assistance is time-consuming and inefficient. A centralized AI-powered system is required to provide instant and reliable responses.

3. Existing System

Traditional systems rely on static websites, notice boards, and manual staff assistance. These methods lack intelligent retrieval and contextual response capabilities.

4. Proposed System

The proposed AI Campus Assistant integrates RAG architecture with Gemini AI. Admin users upload institutional PDFs which are converted into embeddings using Sentence Transformers and indexed with FAISS. When a student asks a question, relevant document chunks are retrieved and passed to Gemini for contextual generation.

5. System Architecture

Frontend (HTML, CSS, JS) → Flask Backend → RAG Retrieval (FAISS) → Gemini AI → Response → Analytics Logging.

6. Technologies Used

- Frontend: HTML5, CSS3, JavaScript
- Backend: Python Flask
- Database: SQLite / PostgreSQL
- AI Model: Google Gemini API
- Vector Database: FAISS
- Embedding Model: Sentence Transformers
- Authentication: Flask-Login
- Security: Flask-Bcrypt

7. Key Features

- Secure Student and Admin Login
- Admin PDF Upload and Deletion
- RAG-Based Document Retrieval
- Multilingual Support (English, Hindi, Kannada)
- Voice Input and Text-to-Speech Output
- Analytics Dashboard with Charts
- Role-Based Access Control

8. Future Enhancements

Future improvements may include cloud deployment, mobile application integration, advanced AI personalization, real-time notifications, and integration with campus ERP systems.

9. Conclusion

AI Campus Assistant demonstrates the effective use of Retrieval-Augmented Generation to create an intelligent academic support system. It enhances accessibility, improves response accuracy, and reduces administrative workload.